

Md Mizanur Rahman (Tanim)

IoT & Full-Stack Developer — Edge AI & GenAI Integration

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Summary

IoT and full-stack developer with many years of challenging experience spanning embedded systems, web platforms, and automation; with a rare crossover into generative AI integration and local/edge AI. Currently a Developer at Wish and Cook in Portugal, building and maintaining the company's IoT gateway and a custom Modbus protocol stack in Python and C, and integrating generative AI into products and internal workflows. Previously ran an independent development practice in Sweden delivering full-stack e-commerce and ERP automation for multiple clients. Passionate about fault-tolerant software, edge computing, local/edge AI, and AI-driven automation. Even though i never get chance to work with Elixir (BEAM VM) but i am very interested and optimistic in learning it and exploring its capabilities for building fault-tolerant and highly concurrent systems. Open to remote or hybrid roles across the EU.

Experience

Developer — Wish and Cook, Ílhavo, Portugal

2023 – Present (permanent employment)

- Build and maintain the company's custom Modbus protocol and gateway written in Python and C with MQTT messaging; architected around a fault-tolerant actor model.
- Write custom applications and scripts for hardware testing and validation — (for rapid iterative development).
- Built a generative-AI pipeline that converts human-written cooking instructions into a machine-readable format, editable in a Next.js-based cookbook.
- Support the team with data analysis, architecture decisions, and programming challenges.

Independent Developer — Sweden (remote)

2018 – 2023

- Built the Nawras Foods e-commerce platform on Shopify with a custom FastAPI/Python backend that automates product sync and generates offers and invoices in Fortnox ERP — (removed manual entry hours for admin).
- Delivered full-stack web and automation projects for multiple clients.

Selected Projects

- **Dispenser Fault-Tolerant App** — fault-tolerant dispenser control application built on an actor-model design. github.com/tanimtbh/dispenser_fault_tolerant_app
- **Hands-On Effective Use of LLMs** — internal workshop materials on applying LLMs to real development tasks. github.com/tanimtbh/hands_on_effective_use_of_llm_workshop
- **Nimai** — Telegram bot built with the OpenAI Agent SDK and the KuzuDB embedded graph database (hobby project). github.com/tanimtbh/nimai
- **Custom Modbus RTU (STM32)** — STM32 firmware project implementing a custom Modbus RTU protocol. github.com/tanimtbh/custom_modbus_rtu

Education

BSc in Electronics and Communication Engineering — Sylhet International University, Sylhet, Bangladesh

2012 – 2016

Graduate Coursework in Intelligent Transport Systems and Logistics — Linköping University, Norrköping, Sweden

2018

Skills

- **Programming Languages:** Python, C, JavaScript
- **AI & Data:** Generative AI integration (OpenAI, Anthropic, DeepInfra, OpenRouter, local models), vector databases (Qdrant, Chroma), graph databases (KuzuDB, Neo4j), data analysis
- **Web & Backend:** React, Next.js, FastAPI, REST APIs, Shopify
- **IoT & Embedded:** Microcontrollers (ESP32 / STM32 / Raspberry Pi), Modbus, MQTT, Edge Computing, Edge AI (Mobilenet v2, Yolo)
- **Tools & DevOps:** Git, GitHub, Linux (bare metal),

Languages

- **English** — Professional working proficiency
- **Bengali (Bangla)** — Native

Availability

Actively seeking remote or hybrid roles to accommodate Portuguese language studies and integration activities.